EXP NO: 11 Experiment to understand pandas library use cases in Data science.

AIM:

To study and experiment with the Pandas library in Python to understand its various use cases in Data Science, including data manipulation, data cleaning, data analysis, and handling structured datasets using Series and DataFrame.

PROGRAM:

```
import numpy as np

import pandas as pd

from google.colab import files


uploaded = files.upload()


list1 = [

   [1, 'Smith', 50000],

   [2, 'Jones', 60000]

]


df = pd.DataFrame(list1)


print("Initial DataFrame:")

print(df)


df.columns = ['Empd', 'Name', 'Salary']


print("\nDataFrame with Column Names:")

print(df)


print("\nDataFrame Information:")

df.info()
```

```
df = pd.read_csv("50_Startups.csv")

print("\nFirst 5 rows of 50_Startups:")
print(df.head())

print("\nLast 5 rows of 50_Startups:")
print(df.tail())

print("\nInformation of 50_Startups:")
df.info()

df = pd.read_csv("employee.csv")

print("\nFirst 5 rows of Employee:")
print(df.head())

print("\nLast 5 rows of Employee:")
print(df.tail())

print("\nEmployee Data Information:")
df.info()

print("\nSalary:")
print(df.salary)

print("\nType of Salary:")
print(type(df.salary))
```

```
print("\nMean Salary:")
print(df.salary.mean())

print("\nMedian Salary:")
print(df.salary.median())

print("\nMode Salary:")
print(df.salary.mode())

print("\nSalary Variance:")
print(df.salary.var())

print("\nSalary Standard Deviation:")
print(df.salary.std())

print("\nDescribe:")
print(df.describe())

print("\nDescribe All:")
print(df.describe(include='all'))

print("\nColumn Names:")
empCol = df.columns
print(empCol)

print("\nNumPy Array:")
emparray = df.values
```

```
print(emparray)


employee_DF = pd.DataFrame(emparray, columns=empCol)


print("\nEmployee DataFrame:")
print(employee_DF)
```

OUTPUT:

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving 50_Startups.csv to 50_Startups (2).csv

Saving employee.csv to employee (1).csv

Initial DataFrame:

** 0 1 2**

0 1 Smith 50000

1 2 Jones 60000

DataFrame with Column Names:

** Empd Name Salary**

0 1 Smith 50000

1 2 Jones 60000

DataFrame Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 2 entries, 0 to 1

Data columns (total 3 columns):

Column Non-Null Count Dtype

--- ------ -------------- -----

```
 0   Empd    2 non-null      int64
 1   Name    2 non-null      object
 2   Salary  2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 180.0+ bytes
```

First 5 rows of 50_Startups:

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

Last 5 rows of 50_Startups:

	R&D Spend	Administration	Marketing Spend	State	Profit
45	1000.23	124153.04	1903.93	California	64926.08
46	1315.46	115816.21	297114.46	Florida	49490.75
47	0.00	135426.92	0.00	California	42559.73
48	542.05	51743.15	0.00	New York	35673.41
49	0.00	116983.80	0.00	California	14681.40

Information of 50_Startups:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 5 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
```

```
 0   R&D Spend        50 non-null     float64
 1   Administration   50 non-null     float64
 2   Marketing Spend  50 non-null     float64
 3   State            50 non-null     object
 4   Profit           50 non-null     float64
dtypes: float64(4), object(1)
memory usage: 2.1+ KB

First 5 rows of Employee:
   emp id                name  salary
0       1     SREE VARSSINI K S    5000
1       2          SREEMATHI B    6000
2       3              SREYA G    7000
3       4  SREYASKARI MULLAPUDI    5000
4       5         SRI AKASH U G    8000

Last 5 rows of Employee:
   emp id                name  salary
2       3              SREYA G    7000
3       4  SREYASKARI MULLAPUDI    5000
4       5         SRI AKASH U G    8000
5       6  SRI HARSHAVARDHANAN R    3000
6       7  SRI HARSHAVARDHANAN R    6000

Employee Data Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 3 columns):
```

```
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   emp id  7 non-null      int64
 1   name    7 non-null      object
 2   salary  7 non-null      int64
dtypes: int64(2), object(1)
memory usage: 300.0+ bytes

Salary:
0    5000
1    6000
2    7000
3    5000
4    8000
5    3000
6    6000
Name: salary, dtype: int64

Type of Salary:
<class 'pandas.core.series.Series'>

Mean Salary:
5714.285714285715

Median Salary:
6000.0

Mode Salary:
```

0 5000

1 6000

Name: salary, dtype: int64

Salary Variance:

2571428.5714285714

Salary Standard Deviation:

1603.5674514745463

Describe:

	emp id	salary
count	7.000000	7.000000
mean	4.000000	5714.285714
std	2.160247	1603.567451
min	1.000000	3000.000000
25%	2.500000	5000.000000
50%	4.000000	6000.000000
75%	5.500000	6500.000000
max	7.000000	8000.000000

Describe All:

	emp id	name	salary
count	7.000000	7	7.000000
unique	NaN	6	NaN
top	NaN	SRI HARSHAVARDHANAN R	NaN
freq	NaN	2	NaN
mean	4.000000	NaN	5714.285714

```
std     2.160247              NaN  1603.567451
min     1.000000              NaN  3000.000000
25%     2.500000              NaN  5000.000000
50%     4.000000              NaN  6000.000000
75%     5.500000              NaN  6500.000000
max     7.000000              NaN  8000.000000

Column Names:
Index(['emp id', 'name', 'salary'], dtype='object')

NumPy Array:
[[1 'SREE VARSSINI K S' 5000]
 [2 'SREEMATHI B' 6000]
 [3 'SREYA G' 7000]
 [4 'SREYASKARI MULLAPUDI' 5000]
 [5 'SRI AKASH U G' 8000]
 [6 'SRI HARSHAVARDHANAN R' 3000]
 [7 'SRI HARSHAVARDHANAN R' 6000]]

Employee DataFrame:
  emp id                name salary
0      1     SREE VARSSINI K S   5000
1      2          SREEMATHI B   6000
2      3              SREYA G   7000
3      4   SREYASKARI MULLAPUDI   5000
4      5         SRI AKASH U G   8000
5      6  SRI HARSHAVARDHANAN R   3000
6      7  SRI HARSHAVARDHANAN R   6000
```

RESULT:

Thus, the experiment was successfully performed, and the various use cases of the Pandas library in Data Science, including data manipulation, data cleaning, data analysis, and handling structured data using Series and DataFrames, were studied and understood.